

## Green Metals

## Battery Metals Watch: Lithium's delayed decline



**Stimulus-led restocking means high lithium prices to remain near term.** In Q2 this year, we called for peaking battery metals prices as mounting supply responses were set to catch up to the accelerating demand. Since then, the cobalt price has fallen 35% and nickel is down 40%, but onshore lithium prices have risen (24%) to record new levels. For lithium, whilst the supply response we had estimated has continued to gather pace, as evidenced by surging Chinese lithium carbonate output, LFP battery expansion related restocking demand and higher-than-expected EV sales in China due to the exceptional stimulus have kept the market tighter than expected. Now, adjusting our global lithium demand assumptions for this stronger restocking environment – which has boosted global lithium consumption by 20% this year – the lithium market is expected to be in an 84kt deficit in 2022 (vs 8kt surplus previously) and a small surplus in 2023 (vs 76kt surplus previously), in our view. However, this merely represents a delay in the swing into surplus and lower prices that we previously forecast for next year. The Chinese EV market is moving towards a demand-constrained state over the next 1-2 years when end-user EV demand will likely lag production, as a stimulus based surge in orders YTD has brought forward much of the demand we expected in 2023-24. Furthermore, we continue to expect strong supply growth in raw materials (brine, spodumene, lepidolite) from Chile, Australia and China over 2023-2025. With downstream overcapacity and slowing EV sales likely to become increasingly apparent over the course of next year, we expect downward pressure on the lithium price to build on surplus cues particularly from H2-23 onward. In this context, we are upgrading our China lithium carbonate spot (excl. VAT) price path through 23H1 to reflect near-term tightness and lagging spodumene contract price pass-through before a decline over the second half of next year to put the 2023 average at \$53,304/t LCE (vs. \$16,372/t LCE previously), though our forecast of a sharp reprice lower in 2024 remains unchanged (GSe \$11,000/t).

**Lithium supply response is gaining momentum.** Just as demand has remained strong, lithium's supply response has also gathered momentum. Chinese chemical production has risen +43% y/y YTD, driven by an increase in raw material supply from both onshore and ex-China assets. Major expansions in China are driven by lepidolite and brine projects on our analysts' estimates and the aggregate run-rate of the majority of projects has reached +89% y/y for 22H2E, up from +56% y/y in 22Q1. Furthermore, exports of Chilean brine have risen by 44kt YTD (6% of global supply),

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whereas 22H1 spodumene production in Australia is up 42% y/y. Moreover, over the last two years, we have seen an increasingly supportive policy environment for electric vehicles from extended subsidies in China to the Inflation Reduction Act in the US. These stimulus measures are now also supporting supply growth through joint ventures between OEMs and junior miners as automakers aggressively pursue securing raw material supply for the EV transition, even as mining capex for base metals remains constrained. We estimate ex-China lithium supply to grow 48% vs. a Chinese supply growth of 37%, which combined will add 221kt LCE of supply over 2023 and then 312kt in 2024.

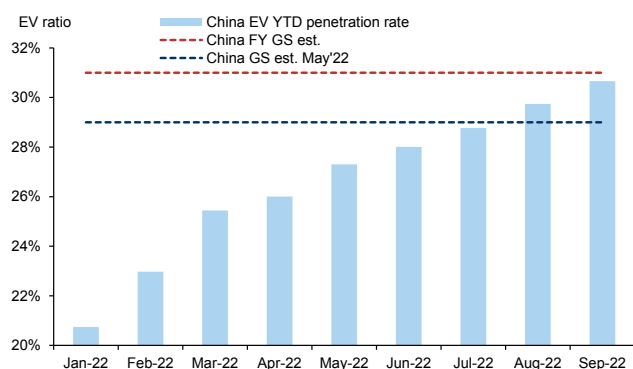
**Decoupling battery supply chain from China is not easy.** Presently China sits at the heart of battery supply chains – from battery materials refining to battery production. The energy crisis has shifted the spotlight on energy security and battery materials supply has become top of mind for policymakers in the West. The lowest-hanging fruit is the lithium supply chain given the lower geographical concentration of the resources compared to cobalt and nickel, in our view. However, c.70% of global refined capacity is in China and this presents a challenge for the West. To compensate for the economies of scale and experience, higher capex flows and regulatory support would be required in order to develop ex-China supply chains. Indeed, the Inflation Reduction Act in the US and the Green Deal in the EU incentivise the OEMs to develop an ex-China battery metals supply chain and help junior miners acquire funding through joint ventures. The task of decoupling from China is most difficult for cobalt which has a high geographical concentration as 80% of smelter supply comes from DRC, whilst refining remains largely located in China.

**Diverging price paths over the next quarters.** Whilst we expect lithium prices to remain supported over 23H1, cobalt and nickel will likely continue on a downward trend. This relative performance is a function of higher share of nickel and cobalt to the European market which faces heightened macro uncertainty into winter as well as better-supplied markets compared to lithium, in our view. We view relative EV sector performance as the best indicator for lithium vs. cobalt demand over the next year. Due to the increased LFP share (56% of total market) in the Chinese EV market, lithium remains most tied to the developments in the onshore EV sector. As a result, we further downgrade our 2023 average cobalt price to \$48,375/t (vs. \$59,500/t prev.) and recently downgraded nickel to \$18,500/t.

# Lithium's delayed decline

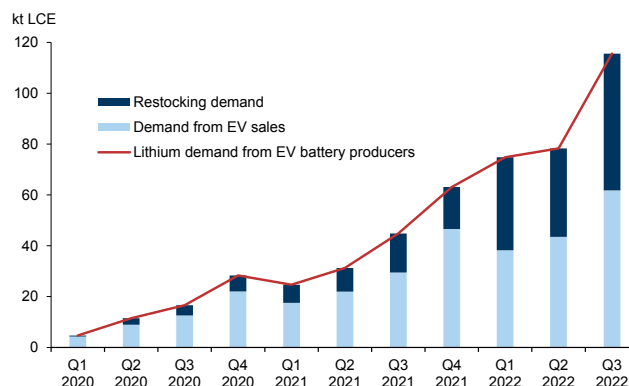
**1. Restocking boosted lithium demand keeps prices high.** The exceptional policy stimulus in China to boost the domestic EV industry has pulled forward battery metals demand this year (see [Exhibit 1](#)). The accelerating EV sales (+110% y/y YTD) combined with the LFP battery capacity expansions in China have added to the sharp rise in lithium demand. As a result of the expansions, we estimate restocking demand globally to have increased to 155kt LCE and has boosted global demand by over 20% this year (see [Exhibit 2](#)). We have significantly adjusted our global lithium demand following the boost from restocking (+74kt LCE), which means the lithium market is now in a 84kt deficit in 2022 (vs 8kt surplus previously) and a small surplus in 2023 (vs 76kt surplus previously). However, the Chinese EV market is moving towards a demand-constrained state over the next 1-2 years as evidenced already by sales underperforming production from Q3 this year (see [Exhibit 4](#)). Finally, we continue to expect strong supply growth (+326kty LCE) in raw materials (brine, spodumene, lepidolite) from Chile, Australia, and China over 2023-2025. We believe the battery maker overcapacity on the back on accelerated capacity build-out amid a decelerating growth of new energy vehicles sales will eventually weigh on lithium prices. We expect this dynamic to start playing out in 2023H2 and a subsequent correction in prices towards the cost curve. In this context, we are upgrading our China lithium carbonate spot (excl. VAT) price path through 23H1 to reflect near-term tightness before a decline over the second half of next year to put the 2023 average at \$53,304/t LCE (vs. \$16,372/t LCE previously), though our forecast of a sharp reprice lower in 2024 remains unchanged (GSe \$11,000/t).

**Exhibit 1: China EV sales have beaten expectations driving lithium demand higher**



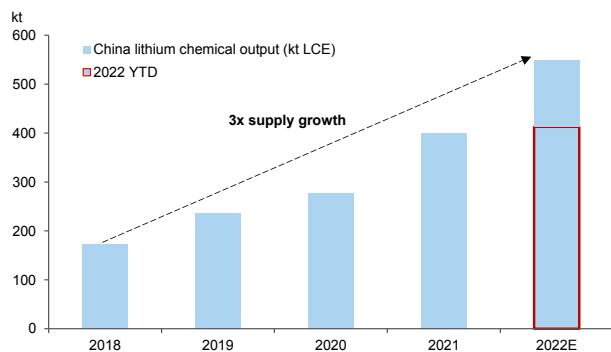
Source: Goldman Sachs Global Investment Research, Wind

**Exhibit 2: An accelerated battery capacity expansion has boosted restocking demand**



Source: Goldman Sachs Global Investment Research, Wind

### Exhibit 3: China's lithium output has risen sharply, in line with our expectations



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research, SMM, Bloomberg, Thomson Reuters

### Exhibit 4: EV retail sales in China have continued to lag production in recent months

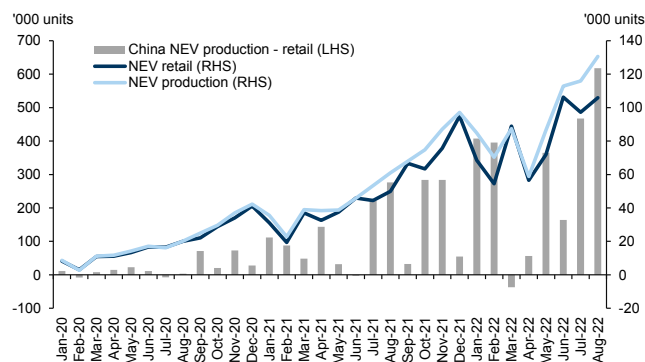
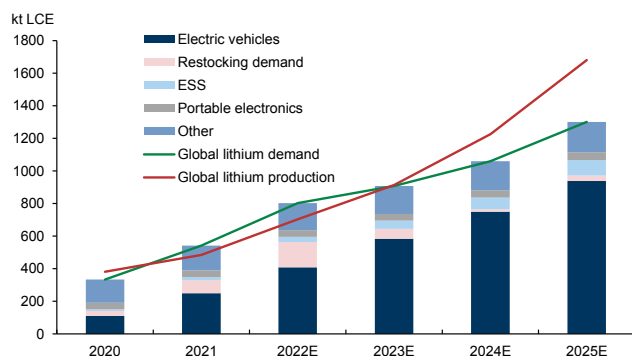


Chart from the "Supply constrained => Demand constrained; Revising our EV market estimates" note

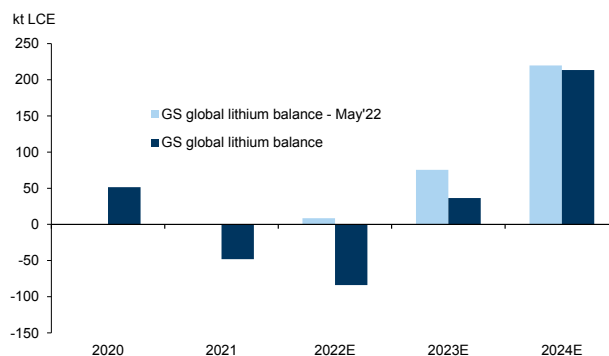
Source: CPCA, Data compiled by Goldman Sachs Global Investment Research

### Exhibit 5: Supply response in the lithium market will outpace the massive demand growth



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research, SMM, Woodmac

### Exhibit 6: High restocking demand has pushed 2022 lithium market into a deficit



Source: Goldman Sachs Global Investment Research, Woodmac, SMM, Data compiled by Goldman Sachs Global Investment Research

Exhibit 7: GS Lithium supply-demand balance

| Li ('000 tonnes LCE)                                         | 2020       | 2021       | 2022E      | 2023E       | 2024E       | 2025E       |
|--------------------------------------------------------------|------------|------------|------------|-------------|-------------|-------------|
| <b>Global demand</b>                                         |            |            |            |             |             |             |
| <b>Consumption - batteries</b>                               | <b>176</b> | <b>326</b> | <b>507</b> | <b>703</b>  | <b>897</b>  | <b>1117</b> |
| % change y/y                                                 | 30%        | 86%        | 55%        | 39%         | 28%         | 24%         |
| EV                                                           | 91         | 215        | 367        | 518         | 664         | 837         |
| ESS                                                          | 11         | 18         | 32         | 51          | 72          | 94          |
| E-buses, two-wheeler EVs                                     | 20         | 34         | 42         | 66          | 86          | 102         |
| Portable electronics                                         | 40         | 42         | 41         | 41          | 44          | 48          |
| Other                                                        | 14         | 17         | 25         | 28          | 32          | 36          |
| <b>Consumption - ex batteries</b>                            | <b>128</b> | <b>135</b> | <b>141</b> | <b>143</b>  | <b>147</b>  | <b>150</b>  |
| % change y/y                                                 | -4%        | 5%         | 5%         | 2%          | 3%          | 2%          |
| Ceramics                                                     | 32         | 33         | 35         | 36          | 37          | 38          |
| Glass-ceramics                                               | 26         | 27         | 29         | 30          | 31          | 31          |
| Other                                                        | 70         | 74         | 77         | 78          | 80          | 81          |
| <b>Restocking demand</b>                                     | <b>30</b>  | <b>81</b>  | <b>155</b> | <b>61</b>   | <b>16</b>   | <b>34</b>   |
| % of global demand                                           |            | 17%        | 24%        | 7%          | 2%          | 3%          |
| <b>Global demand (excl. restocking)</b>                      | <b>304</b> | <b>461</b> | <b>648</b> | <b>847</b>  | <b>1044</b> | <b>1267</b> |
| <b>Global demand (incl. restocking)</b>                      | <b>334</b> | <b>542</b> | <b>803</b> | <b>908</b>  | <b>1060</b> | <b>1301</b> |
| % change y/y                                                 | 24%        | 62%        | 48%        | 13%         | 17%         | 23%         |
| <b>Global Refined Supply</b>                                 |            |            |            |             |             |             |
| <b>Brine</b>                                                 | <b>168</b> | <b>223</b> | <b>305</b> | <b>411</b>  | <b>506</b>  | <b>618</b>  |
| China                                                        | 31         | 55         | 75         | 99          | 133         | 194         |
| Ex-China                                                     | 137        | 168        | 229        | 311         | 372         | 424         |
| <b>Spodumene</b>                                             | <b>189</b> | <b>233</b> | <b>320</b> | <b>491</b>  | <b>726</b>  | <b>1011</b> |
| China                                                        | 10         | 7          | 15         | 15          | 38          | 84          |
| Ex-China                                                     | 179        | 225        | 305        | 476         | 687         | 927         |
| <b>Other</b>                                                 | <b>25</b>  | <b>49</b>  | <b>97</b>  | <b>142</b>  | <b>211</b>  | <b>349</b>  |
| China                                                        | 25         | 49         | 97         | 142         | 211         | 331         |
| Ex-China                                                     | 0          | 0          | 0          | 0           | 0           | 18          |
| <b>World output</b>                                          | <b>382</b> | <b>505</b> | <b>721</b> | <b>1044</b> | <b>1442</b> | <b>1977</b> |
| % change y/y                                                 |            | 32%        | 43%        | 45%         | 38%         | 37%         |
| <b>Total output (adj. for disruption)</b>                    | <b>382</b> | <b>485</b> | <b>703</b> | <b>913</b>  | <b>1226</b> | <b>1681</b> |
| % change y/y                                                 |            | 27%        | 45%        | 30%         | 34%         | 37%         |
| <b>Battery Scrap Supply</b>                                  | <b>4</b>   | <b>9</b>   | <b>16</b>  | <b>31</b>   | <b>47</b>   | <b>71</b>   |
| % change y/y                                                 |            | 2%         | 2%         | 3%          | 4%          | 4%          |
| <b>Global Balance</b>                                        | <b>52</b>  | <b>-48</b> | <b>-84</b> | <b>37</b>   | <b>213</b>  | <b>450</b>  |
| as % of global supply                                        | 14%        | -10%       | -12%       | 4%          | 17%         | 27%         |
| <b>Cash Prices (Li carbonate China spot, annual average)</b> |            |            |            |             |             |             |
| Current dollar (excl. VAT \$/tonne)                          | 5500       | 15656      | 62929      | 53304       | 11000       | 11000       |
| Current yuan (CNY/tonne)                                     | 42863      | 120544     | 483990     | 430833      | 68000       | 68000       |

Price refers to Fastmarket lithium carbonate China spot (Yuan/tonne) and the dollar/tonne price excluded 13% VAT post conversion

Source: Goldman Sachs Global Investment Research, Wind, BNEF, Woodmac, SMM, Data compiled by Goldman Sachs Global Investment Research

**2. Policy continues to dictate demand.** Government policies, stimulus measures and geopolitics of energy security have continued to change the landscape for electric vehicles globally. Whilst a bigger than expected stimulus has boosted Chinese EV sales and the Inflation Reduction Act (IRA) presents a positive US EV outlook, the persistent inflation and cost of living crisis has lowered EV sales growth estimate in Europe over the next year. The Chinese government has spent RMB140bn (or c.\$20bn) on the auto industry – 60% of which has favoured EVs – frontloading demand to 2022. As a result, our auto analysts now expect Chinese EV sales to rise c.6mil (vs. 5.2Mil prev.) in 2022 and increase to 7.1Mil in 2023 (vs. 6.2Mil prev). In Europe, the declining consumer

confidence amid high inflation and the energy crisis has decreased available cashflow to households which is expected to trough in 23H1. However, the pent-up demand and already low auto volumes reduce downside risk to 2023 volumes. In this context, the European auto team sees EV sales to amount to 3.1Mil (vs. 4.5Mil prev.) in 2023. Finally, we believe the IRA will support the long-term EV adoption, increase investments in the battery supply chains and push US OEMs to go all EV in the long-term. As a result of revisions to our global EV sales models, we now expect lithium demand from batteries to grow 39% in 2023, nickel at 28% and cobalt at 19%. The divergence in demand comes from the increased leverage of lithium to the Chinese EV market in the form of LFP dominance and relatively weaker growth in the EU which favours NMC batteries for now.

**Exhibit 8: The orderbooks in Europe have started to decline as a result of the cost-of-living crisis**

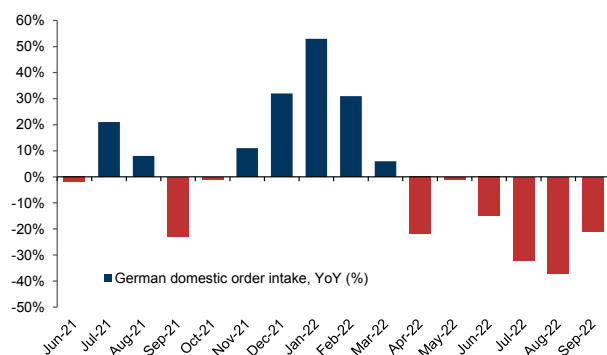
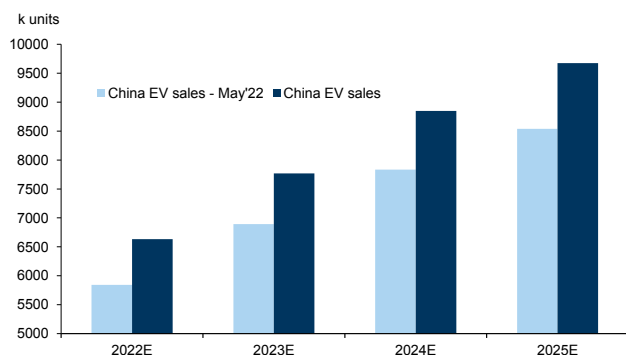


Chart from the European Auto team's linked note

Source: Goldman Sachs Global Investment Research, VDA

**Exhibit 10: Continuation of incentives has increased China EV sales GSe by 1.6mn in 2022/23**

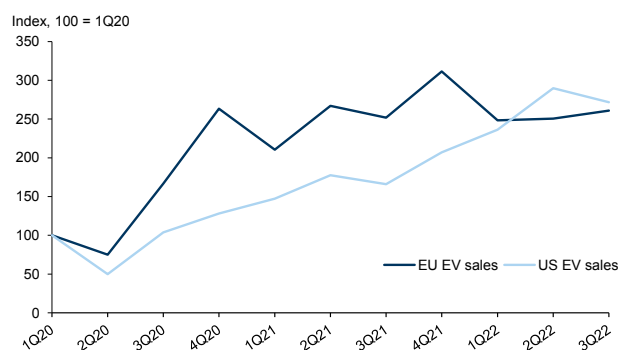
EV sales include PHEV and BEV



Source: Goldman Sachs Global Investment Research

**Exhibit 9: Rising inflation and recessionary environment has weighed on EU EV sales**

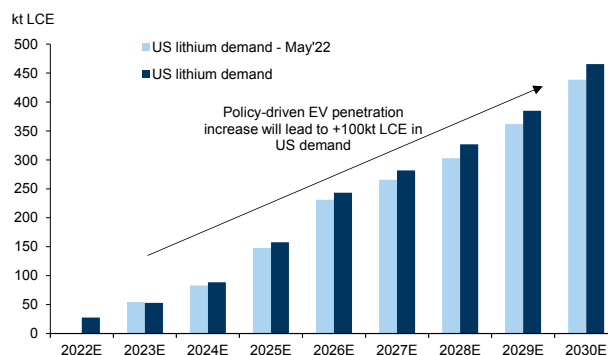
EV sales include PHEV and BEV, indexed at 1Q22



Source: Wind, Goldman Sachs Global Investment Research

**Exhibit 11: Inflation Reduction Act to sustain US EV penetration and lithium demand**

Based on GS model estimates

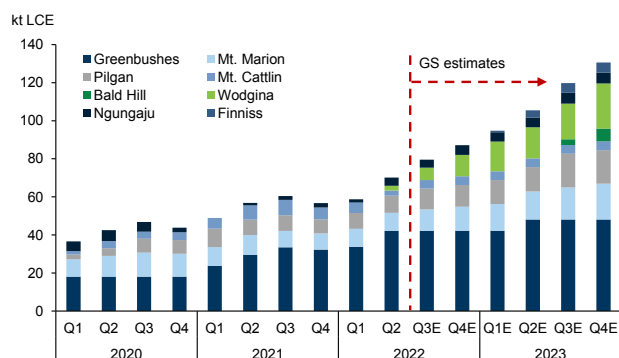


Source: Goldman Sachs Global Investment Research

**3. Supply response is gaining momentum.** For lithium, we have seen a strong increase in Chinese chemical production (+43%y/y YTD) driven by an increase in raw material supply from both onshore and ex-China assets. We have noted this earlier and as our equity analysts detail below, major expansions in China are driven by lepidolite

and brine projects on our estimates. The aggregate run-rate of the majority of projects has reached 89% of 22H2E, up from 56% in 22Q1. Furthermore, exports of Chilean lithium compounds have risen by 44kt YTD (6% of global supply) whereas 22H1 spodumene production in Australia is up 42% y/y. The significant supply growth phase has been catalysed by joint ventures between OEMs and junior miners as automakers aggressively pursue securing raw material supply for the EV supply chain, even as mining capex for base metals remains constrained. In Australia, we continue to see a ramp-up on upstream as well as an acceleration to produce hydroxide. We estimate ex-China lithium supply to grow 42% vs. Chinese supply growth of 37% and combined will add 221kt LCE of supply over 2023. On nickel, we expect battery grade chemical output to increase by 117kt as Indonesian intermediate supply grows in 2023. Whilst intermediate production and exports from Indonesia have risen this year (+240% y/y), there exists a shortage of smelter processing capacity to convert matte and MHP units into sulphate for use in the battery sector. We expect this bottleneck will likely be solved over the next 12 months. Finally, we continue to expect cobalt supply to benefit from the phase of peaking copper mine supply and Indonesian HPAL projects as a key by-product over the next 12-18 months.

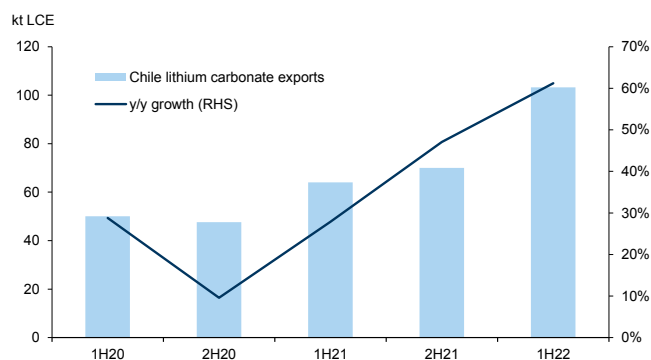
**Exhibit 12: Wodgina and Pilgan assets to add significant lithium supply out of Australia**



Source: Goldman Sachs Global Investment Research, Company data, Goldman Sachs Global Investment Research

**Exhibit 13: Rising Chilean exports (+60%y/y YTD) have added to global supply growth YTD**

Chile lithium carbonate exports



Source: Comtrade, Goldman Sachs Global Investment Research

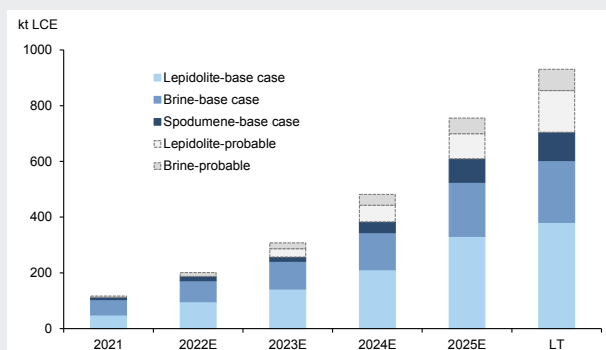
## An update on global lithium supply

### China supply (Trina Chen / Joy Zhang)

Based on our bottom-up work, we estimate total development projects (base case, probables and others) reached 34 as of the end of September, versus 24 in June. As a result, we now see 31-42% higher expansions in 2024-25, while our estimate for 2022-23 remains mostly unchanged. Specifically, we expect China domestic production to deliver 76kt-LCE expansions in 2022 and 70-226 kt-LCE in 2023-2025.

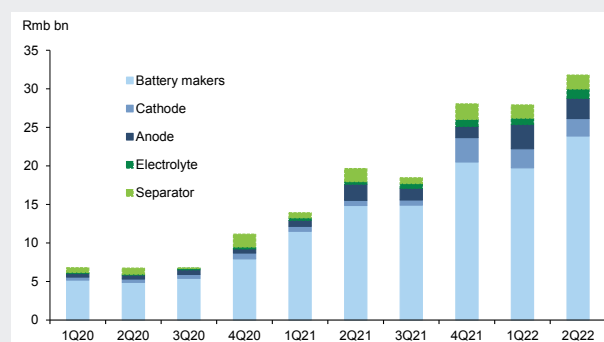
#### Exhibit 14: China lithium chemical supply by upstream resources - 2021-2025E

Annual output of lithium carbonate (upstream implied) - China (kt LCE)



Source: Wood Mackenzie, Company data, Goldman Sachs Global Investment Research

#### Exhibit 15: Aggregated CAPEX of battery supply chain by key producers

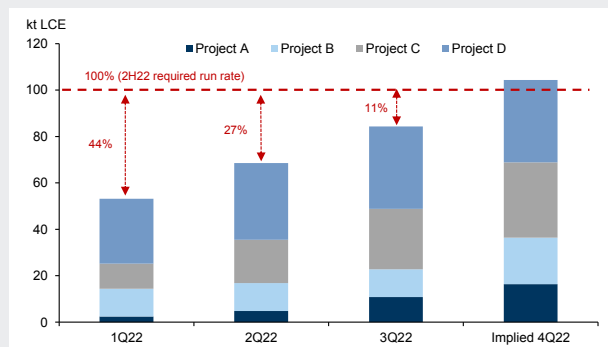


Source: Company data, Goldman Sachs Global Investment Research

Major expansion projects are driven by domestic lepidolite and brine projects, with much improved economics due to optimizing process technologies (see [link](#)). We have been tracking key expansion projects on the ground, and have visited Jiangxi Yichun, where the majority of lepidolite mines and chemicals projects are being developed. We see challenges in the expansions including the approval for a beneficiary and tailing dam, and a potential bottleneck on local transport infrastructure. Nevertheless, our tracking of selected projects also suggests the project ramp-ups are mostly in line with the expectations YTD 2022. The aggregate run-rate of the selected projects has reached 89% of the expected 2H22E, from 56% in 1Q22, and 73% in 2Q22, on our estimates.

**Exhibit 16: Annualized run-rate for selected lithium upstream projects**

Annualized run-rate for selected lithium upstream projects (kt LCE)



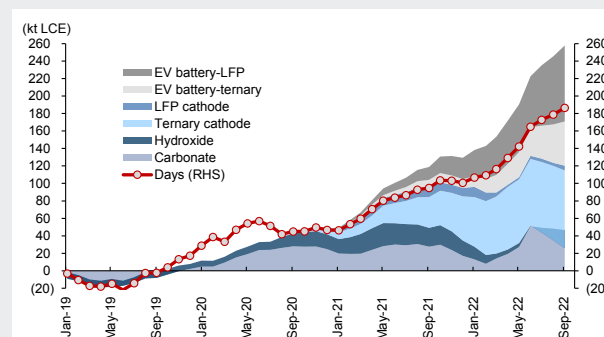
Source: Company data, Goldman Sachs Global Investment Research

**Exhibit 17: Mountain view - Yichun, Jiangxi**

Source: Goldman Sachs Global Investment Research

**Exhibit 18: Mountain view - Yichun, Jiangxi**

Source: Goldman Sachs Global Investment Research

**Exhibit 19: Lithium inventory (implied) across EV supply chain**  
Implied inventory in equivalent Li<sub>2</sub>CO<sub>3</sub> (kt LCE) / days

Source: SMM, Goldman Sachs Global Investment Research

We highlight key observations and takeaways on selected projects we track:

- **Project A:** located in Yichun, Jiangxi, the project is an integrated project from local lepidolite to mica, and lithium carbonate. The project targets for 8kt LCE Lithium carbonate output for 2022E, 20kt LCE for 2023E, and 80-120kt LCE by 2025E-LT. The 1st mine acquired by the company has obtained the mining license and already in operation. As of September 2022, the project was running at 10.8kt LCE annualized run-rate, 79% of our estimated 2H22E run-rate, vs 54% in 2023E and 14% in 2025E. The project is still waiting for two other mining licenses for two additional mining areas, and likely to obtain the licenses in October or November. The current production costs are around Rmb120,000/LCE for final chemical products. With the project ramping up, it expects production costs to normalize to local average level.
- **Project B:** located in Yichun, Jiangxi, the project is an integrated lepidolite project with one operating mine in expansion and three mines waiting for mining licenses. The project currently runs at 12kt LCE annualized run-rate, 75% of our estimated 2H22E run-rate. Assuming the mining licenses obtained in 2H23E, the project could achieve 30kt LCE output in 2023E, and target to reach 40-50kt LCE by 2025E-LT. The current production costs (excl VAT) for the project are around Rmb70,000/t LCE, of which

mining costs composing Rmb30,000/t LCE.

- Project C: located in Yichun, Jiangxi, the integrated lepidolite project renewed its mining license in April 2022 with the approved capacity increasing by 3x size of original one at 1,000kt/year. At the same time, the project has expanded its conversion capacity from 10kt LCE to 30kt LCE in 1H22. As the expansion is ramping up, the project has reached 26kt LCE annualized run-rate in 3Q22, 89% of our estimated 2H22E run-rate. The project targets for 22kt LCE lithium carbonate output in 2022E, 30kt LCE in 2023E and 50kt LCE in 2025E-LT with the construction of phase III. Due to the increasing energy prices, the production cost has increased by 16% to Rmb50,000/t LCE from the 2021 level.
- Project D: located in Qinghai province, the project is a brine project with 30kt/year capacity in operation. Among which, 10kt/year capacity has been running stable since 2017. Due to the commercial operation of its 20kt/year new capacity in Jun 2022, the project targets to achieve 33kt LCE lithium carbonate output in 2022E, and 36-37kt LCE in 2023E through improvement of extraction efficiency. As of September 2022, the project was running at 35.5kt LCE annualized runrate, 100% of our estimated 2H22E run-rate. The brine also has two ongoing expansion projects with annual capacity of 30kt LCE and 40kt LCE respectively. Both projects will adopt similar technology route as the current operating one. With the two expansions, the project targets 107kt LCE lithium carbonate output in 2025E and 117kt LCE in LT. The production cost remains stable at around Rmb28,000/t LCE.

The supply expansion along the China battery supply chain is also leading to a stronger-than-expected inventory increase as producers prepare the working stock. Based on our estimates of industry inventory across the supply chain, total inventory, in the form of battery, cathodes and chemicals etc, in equivalent lithium carbonate, reached ~260kt LCE, as of the end of September 2022. This is equivalent to 186 days and 107 days of 2023E China and global demand, respectively. Compared to 223kt LCE as of June 2022, the implied inventory restocking of 35 kt LCE is equivalent to 28% of Chinese lithium demand (based on the 2023E run-rate). The aggressive restocking is consistent with the strong capex uptrend in the industry — the aggregated quarterly capex by major Chinese producers across the supply chain reached Rmb30bn by 2Q22A, 60% higher than the average of 2021A.

### Ex-China supply (Paul Young / Hugo Nicolaci)

The Australian upstream producers continue to ramp up and grow production, with lithium units currently being prioritised over grade in the current high pricing environment, while work continues to be done to accelerate the move downstream into hydroxide by the larger producers. A summary of the state of the key Australian upstream producers/projects is summarised below.

- Talison's (Tianqi/Albemarle/IGO) **Greenbushes** (~1,100kt spodumene concentrate production in FY22) continues to grow production with the ramp up of the tailings retreatment project, while chemical grade plants 3 & 4 potentially add an incremental 520ktpa of capacity each from mid-late decade.
- Pilbara Minerals' **Pilgangoora** is growing production to 540-580kt in FY23, and ramping up to 680ktpa from 4QCY23 following FID this year on a 100ktpa expansion. Current unallocated volumes are being sold via the BMX auction platform, and the business plans to FID a further expansion to 1,000ktpa by the end of 2022 (with additional volumes currently unallocated).
- At **Wodgina** (Mineral Resources/Albemarle), trains 1 & 2 (250ktpa 6% spodumene nameplate) started

in March & July 2022 respectively, with train 3 (additional 250ktpa) scheduled to start production in mid-2023.

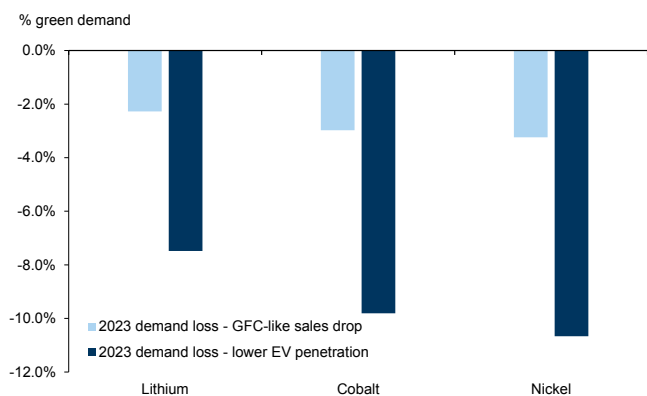
- At **Mt. Marion** (Mineral Resources/Ganfeng) is targeting completion of the planned two stage upgrade to ~600ktpa 6% spodumene (900ktpa mixed grade) by year-end.
- Allkem's **Mt Cattlin** spodumene project will have rebounding volumes through FY23, impacted through 2H22 by lower ore grades/recoveries. Construction activities at **James Bay** are targeted to commence in early 2023 with commissioning of the 320ktpa spodumene project in mid-2024. On brine assets, **Olaroz Stage 2** has been delayed on equipment availability with production now expected from 2Q 2023, taking Olaroz to ~40ktpa of lithium carbonate by mid-decade, and at **Sal de Vida** the two-stage 45ktpa carbonate project is under construction with first production from the 15ktpa Stage 1 expected by H2 CY23 (development of Stage 2 to commence in parallel with Stage 1 commissioning).
- Core Lithium's **Finniss** mine is in ramp up, targeting first DSO sales by December 2022 and spodumene concentrate production in 1H23, ramping up to ~180ktpa from 2024, with ~80% of production over the first four years sold to offtake partners (including Ganfeng Lithium and Sichuan Yahua). The company is looking to further resource growth across its exploration portfolio, with possible extensions at BP33 underground.
- Liontown's **Kathleen Valley** received final key approvals to enable project construction and operations in early October 2022 (following FID in June), with contracts for all key long-lead items already executed and in manufacture or ready for shipment, and construction expected to ramp up over the remainder of 2022 and into 2023. Completion and commissioning is targeted by the end of Q2 2024 for the initial project aiming to deliver 500ktpa of spodumene concentrate, with a planned expansion to 700ktpa of spodumene concentrate within six years, and offtakes in place for at least the first 5 years with LG Energy, Tesla and Ford covering ~90% of production.
- Covalent's (SQM/Wesfarmers) **Mt. Holland** project is progressing, targeting production from 4Q 2023 and ramping up to 300ktpa of spodumene concentrate.
- Rio Tinto's **Rincon** is progressing early works including construction of a camp and airstrip, and developing a 3ktpa lithium carbonate starter plant (not in GS base case) to accelerate market entry by 1H 2024, and de-risk the planned full scale operation. Detailed studies for the full-scale operation are progressing. At **Jadar** Rio is exploring all options, acknowledging the concerns from local communities and is committed to meaningful engagement to explore ways to address these.

*Note: The above discussion of China's lithium supply sector represents the views of Trina Chen and Joy Zhang who are equity analysts covering China Basic Materials and ex-China supply views are attributed to Paul Young and Hugo Nicolaci who are equity analysts covering Australia Basic Resources. All other views are those of the Commodities research team.*

**4. Upside risk to lithium prices limited to restocking demand.** Lithium fundamentals have sequentially beaten expectations over the past two years due to the continued growth in the EV sector, particularly in China. We expect 2023 to follow a similar path and expect battery-related demand to grow 39% whilst supply rises 36% y/y. Yet

uncertainties remain elevated - macroeconomic environment in Europe could further depress sales or a slower ramp-up of production in China amid Covid lockdowns could suppress supply. However, further material downgrades to Europe demand remain limited by strong pent-up demand due to 2020-21 supply chain constraints. As such, we expect any further disappointment in EV sales in Europe to have a marginal impact on global lithium demand. In a scenario of a GFC-like drop in 2023 European auto sales (-8% y/y), we estimate the lithium demand loss would only amount to 14kt (2% of global green demand). The impact of such a scenario will be higher on cobalt (-2kt, 3% of global demand) and nickel (-13kt, 3.2% of global demand) markets given the high share of NMC Europe. Higher uncertainty remains on restocking demand, in our view. We currently expect 2023 restocking demand to be driven by working stocks requirements, with no expected growth for stocks related to battery capacity expansion, opposed to the 2022 exceptionally high growth. Yet, we note that any continued restocking in China at new battery plants or a pickup in ex-China restocking could push the market into a deficit, impacting the pace of the downward trend in prices (see [Exhibit 21](#)). We estimate that a repeat of 2022 restocking demand would move 2023 market from a modest surplus (37kt LCE) to a deficit (61kt LCE), while a restocking to 15 weeks of demand (half of 2022), would essentially balance the market (15kt deficit). Finally, in the mid-term the restocking path will not have a material impact on balances and prices, given the sizes of the expected surpluses starting from 2024-25, in our view.

**Exhibit 20: Pent-up demand and already “recession-like” volumes limit EU auto sales downside**



Source: Goldman Sachs Global Investment Research

**Exhibit 21: A continuation of restocking demand into 2023 can keep lithium prices high**

Global lithium balance under different scenario of restocking needs for battery capacity expansions



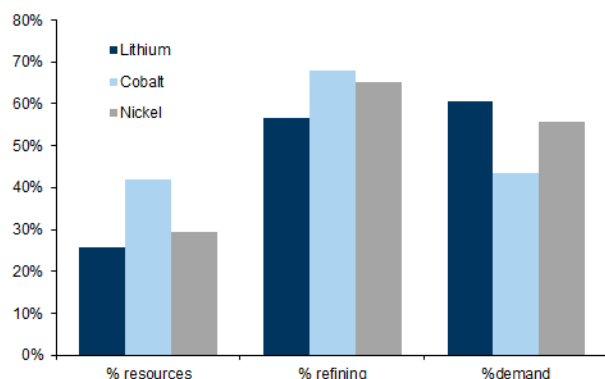
Source: Goldman Sachs Global Investment Research

**5. Deglobalisation of battery materials supply chains is not easy.** High energy prices and gas scarcity in Europe have put a spotlight of the global battery supply chain. Batteries are central to the future of mobility, energy security and the green transition, with EV penetration rates now expected to reach 82% and 70% in EU and US respectively. As a result, policymakers and OEMs have started to invest in relocation of supply chains closer to home. In our view, lithium supply chain is the low-hanging fruit in terms of relocation, compared to more concentrated and mature markets for nickel and cobalt (see [Exhibit 22](#)). For lithium, whilst refining is highly concentrated in China (c.70% of global capacity), only 26% of hard rock lithium and brine comes from there. That being said, decoupling from China remains a challenge given the economics of scale

and relative gap in experience. To compensate for these factors, higher capex flows and regulatory support would be required in the Western world. Indeed, the Inflation Reduction Act and Bipartisan Infrastructure funding in the US and the Green Deal in the EU incentivises the OEMs to source battery metals for the green transition and helps junior miners through joint ventures (see Exhibit 24). The task of decoupling from China is most difficult for Cobalt which has a high geographical concentration as 80% of smelter supply comes from DRC, whilst refining remains largely located in China.

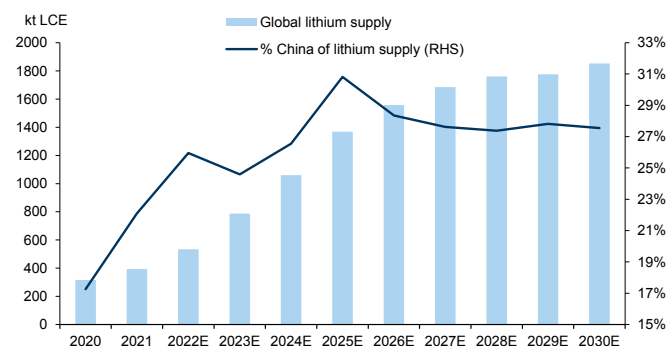
#### Exhibit 22: Refining for battery metals is heavily concentrated in China

Chinese share of upstream and refining assets and global demand share



Source: Woodmac, Goldman Sachs Global Investment Research

#### Exhibit 23: China only accounts for one-third of lithium raw material supply



Source: Woodmac, Goldman Sachs Global Investment Research

#### Exhibit 24: OEMs are increasingly signing deals with ex-China lithium producers

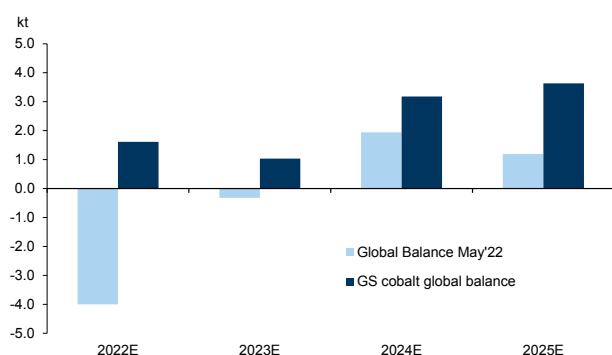
| Recent ex-China Lithium sourcing/Joint venture announcements |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tesla</b>                                                 | - Liontown Resources to supply spodumene concentrate for 5 years starting from 2024. Supply is expected at 100kt in the first year and 150kt in years 2-5<br>- Extended lithium contract with Livent through 2021 and is working on a long term deal                                                                                                                                                           |
| <b>Volkswagen</b>                                            | - Vulcan Energy Resources will provide lithium hydroxide for five years starting 2026                                                                                                                                                                                                                                                                                                                          |
| <b>Stellantis</b>                                            | - Investment in Vulcan Energy Resources for lithium supply agreement<br>- Controlled Thermal Resources to supply up to 25kt/y of lithium hydroxide over 10 years from a project in California                                                                                                                                                                                                                  |
| <b>GM</b>                                                    | - Prepayment to Livent for a guaranteed six year supply of lithium from 2025 at a contractual price per tonne<br>- Invest and help develop Controlled Thermal Resources Ltd's geothermal brine project near California's Salton Sea (60kt of potential production)                                                                                                                                             |
| <b>Ford</b>                                                  | - Ioneer Ltd signs binding offtake agreement to supply lithium from Rhyolite Ridge in Nevada<br>- Agreement with Compas Materials for lithium hydroxide and lithium carbonate in Utah<br>- Liontown Resources to supply up to 150k dmt/y of spodumene concentrate from its Kathleen Valley project for five years starting 2024<br>- Preliminary deal to purchase 25kt of lithium annually from Lake Resources |
| <b>BMW</b>                                                   | - Partnership with Livent and Ganfeng to acquire lithium supply                                                                                                                                                                                                                                                                                                                                                |
| <b>Mercedes</b>                                              | - Supply agreement with Rock Tech Lithium for 10kty battery-grade lithium supply                                                                                                                                                                                                                                                                                                                               |

Source: Data compiled by Goldman Sachs Global Investment Research, Thomson Reuters, Company data

**6. Cobalt – further fall in prices from demand pressures.** Cobalt prices have corrected by 35% since May, touching a low of \$52,800/t in August. This is reflective of a continuous deterioration in cobalt demand globally, in our view. Indeed, due to lower EV sales in Europe, an increasing shift to low-cobalt NMC811/955 and rapid growth of LFP in China, alongside underperformance of the electronics sector, we now expect 2022 cobalt demand to grow by 7% (vs 17% previously). The downgrade in EV battery demand accounts for 80% of the total decrease (18kt), with portable electronics slowdown accounting for the rest. Our auto teams expect relatively lower EV sales in Europe and higher LFP share in 2023; therefore we downgrade cobalt battery demand by 16kt to 132kt (vs. 111kt this year). The 2022-23 demand downward adjustment is only partially offset by slower-than-expected growth in DRC production (-8kt) and continued disruptions along the exports route from the country due to floods earlier in the year and

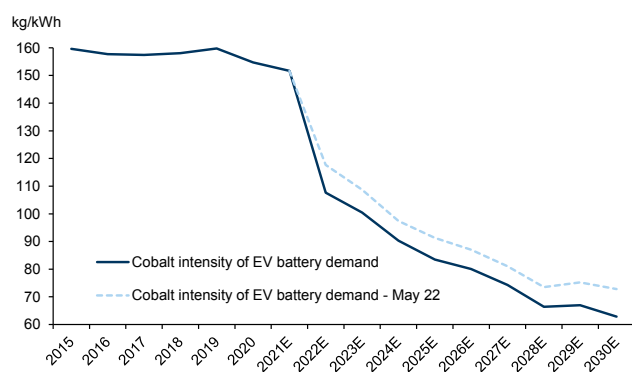
a ban on some volumes since July. Moreover, continued delays in exports have led to a build-up in stocks in the DRC, adding to the expected strong growth in the country's cobalt into 2023 (+9% y/y) which is a consequence of the strong copper mine production (+9% y/y). Finally, Indonesian nickel-cobalt MHP is emerging as an alternative for DRC hydroxide in NMC batteries due to the already existing nickel content. As such, we expect the market to turn into a small surplus in 2022, at 1.6kt and to essentially balance into 2023 at 1kt. Moreover, the risk to demand is skewed to the downside if the LFP share continued to expand outside China, especially in Europe. In this context, we downgrade 2023 and 2024 annual price average at \$48,375 and \$44,000/t (vs previously \$59,500/48,000/t).

**Exhibit 25: Cobalt market will move into surplus in 2022 as demand remains weak**



Source: Woodmac, BNEF, Fastmarkets, Goldman Sachs Global Investment Research

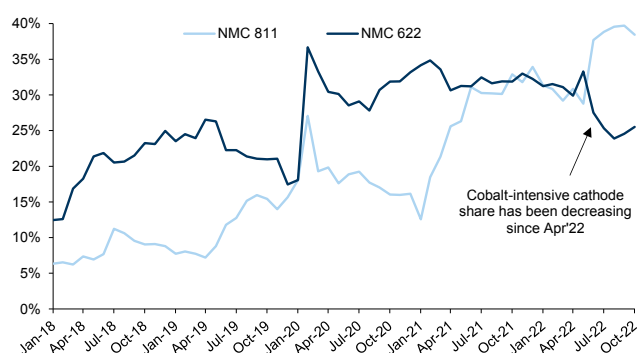
**Exhibit 27: Lower European EV sales and higher LFP share have contributed to declining EV demand cobalt intensity**



Source: BNEF, Woodmac, Goldman Sachs Global Investment Research

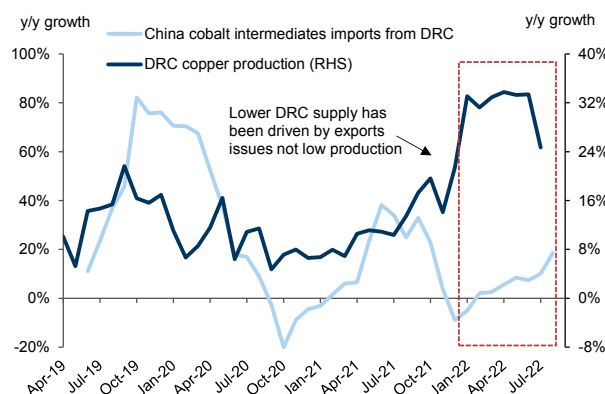
**Exhibit 26: Sequential shift to low cobalt NMC 811 has lowered demand from EVs**

NMC 811 & NMC 622 share of ternary precursor output



Source: Wind, Goldman Sachs Global Investment Research

**Exhibit 28: Lower exports from DRC this year has increased 2023 expected supply**



Source: SMM, ICSG, Goldman Sachs Global Investment Research

Exhibit 29: GS Cobalt supply-demand balance

| ('000 tonnes)                             | 2020       | 2021       | 2022E      | 2023E       | 2024E      | 2025E      |
|-------------------------------------------|------------|------------|------------|-------------|------------|------------|
| <b>Global demand</b>                      |            |            |            |             |            |            |
| <b>Consumption - batteries</b>            | <b>73</b>  | <b>102</b> | <b>111</b> | <b>132</b>  | <b>149</b> | <b>168</b> |
| % change y/y                              | -4%        | 40%        | 8%         | 19%         | 13%        | 13%        |
| EV                                        | 19         | 44         | 53         | 70          | 82         | 96         |
| ESS                                       | 2          | 2          | 3          | 4           | 6          | 7          |
| E-buses, two-wheeler EVs                  | 1.9        | 2.8        | 2.0        | 3.0         | 4.7        | 5.2        |
| Portable electronics                      | 47         | 50         | 49         | 50          | 52         | 55         |
| Other                                     | 4          | 4          | 4          | 4           | 4          | 4          |
| <b>Consumption - ex batteries</b>         | <b>53</b>  | <b>57</b>  | <b>59</b>  | <b>60</b>   | <b>63</b>  | <b>65</b>  |
| % change y/y                              | -14%       | 8%         | 3%         | 3%          | 5%         | 4%         |
| Superalloys                               | 13         | 13.7       | 13.5       | 14.2        | 15         | 15         |
| Tool materials                            | 11         | 12.0       | 12.3       | 12.9        | 13         | 14         |
| Other                                     | 29         | 31.1       | 32.9       | 33.0        | 35         | 36         |
| <b>Global consumption</b>                 | <b>125</b> | <b>159</b> | <b>169</b> | <b>192</b>  | <b>212</b> | <b>233</b> |
| % change y/y                              | -9%        | 27%        | 7%         | 13%         | 10%        | 10%        |
| <b>Global Production</b>                  |            |            |            |             |            |            |
| <b>World smelter output</b>               | <b>138</b> | <b>154</b> | <b>187</b> | <b>205</b>  | <b>227</b> | <b>250</b> |
| % change y/y                              | -5%        | 12%        | 21%        | 10%         | 10%        | 10%        |
| DRC                                       | 96         | 113        | 141        | 154         | 170        | 189        |
| Ex- DRC                                   | 42         | 42         | 46         | 51          | 57         | 60         |
| <b>World refined output</b>               | <b>133</b> | <b>149</b> | <b>181</b> | <b>198</b>  | <b>219</b> | <b>241</b> |
| <b>World output (adj. for disruption)</b> | <b>133</b> | <b>149</b> | <b>166</b> | <b>185</b>  | <b>205</b> | <b>222</b> |
| % change y/y                              |            | 12%        | 12%        | 12%         | 10%        | 8%         |
| <b>Battery Scrap Supply</b>               | <b>3</b>   | <b>4</b>   | <b>5</b>   | <b>7</b>    | <b>10</b>  | <b>15</b>  |
| % change y/y                              |            | 3%         | 3%         | 4%          | 5%         | 7%         |
| <b>Global Balance</b>                     | <b>11</b>  | <b>-6</b>  | <b>1.6</b> | <b>1.03</b> | <b>3.2</b> | <b>3.6</b> |
| as % of global supply                     | 8%         | -4%        | 1%         | 1%          | 2%         | 2%         |
| <b>Cash Prices (annual average)</b>       |            |            |            |             |            |            |
| Current dollar (\$/t)                     | 33947      | 51119      | 69100      | 48375       | 44000      | 44000      |
| Current dollar (c/lb)                     | 1540       | 2319       | 3134       | 2194        | 1996       | 1996       |

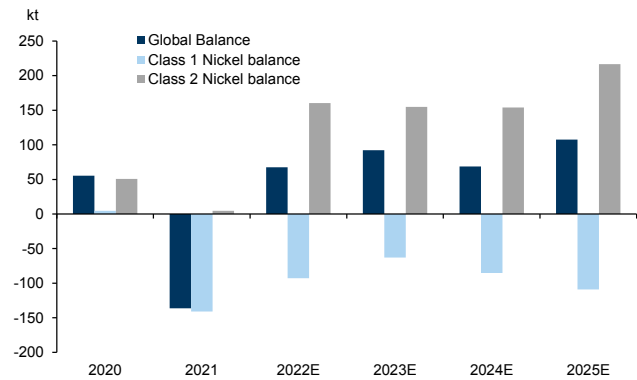
Prices refer to Fastmarket cobalt metal standard grade

Source: Woodmac, BNEF, Goldman Sachs Global Investment Research

**7. Nickel – Ex-battery market moving to glut, policy risk premium for now.** The energy crisis in the EU has created macro headwinds in the region amid worsening cost-of-living crisis. Our EU auto analysts expect lower EV sales in 2022 and 2023 (2.1mil EV sales in 2023 vs. 3.2mil EV sales previously). Furthermore, the aggressive destocking and end-demand softening in Europe over uncertain demand conditions in winter has induced a negative shock to the class 2 nickel market, with softening repercussions to the broader balance. We estimate a combined 64kt demand hit in 2022-23 from the EU's weak stainless sector and on the supply side, we have seen strong supply growth from Indonesia as it ramps up both NPI and intermediates capacity. On the supply side, we expect battery grade chemical output to increase by 117kt as Indonesian intermediate capacity grows in 2023 and see an additional 74kty of NPI capacity additions on average each year over the next 3 years. As a result, we estimate the global balance to be in a 68kt surplus for this year and a 92kt surplus for next year. However, the battery grade class 1 nickel market remains relatively tight with LME stocks now at their lowest level since the early 2000s. This remains the case because even though intermediate production and exports from Indonesia have risen (+240% y/y), there exists a shortage of smelter processing capacity to convert matte and MHP units into sulphate for use in the battery sector. We expect this bottleneck will

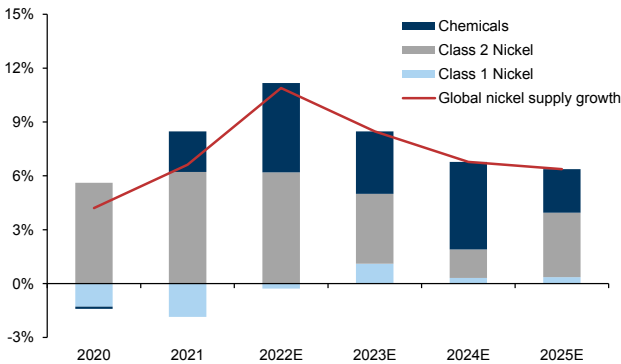
likely be solved over the next 12 months but for now it is a cause of tightness in the class 1 market. In this context, it is also important to recognise that Russia is a significant supplier of class 1 nickel (17% global supply, 7% total nickel supply) which in the near term means policy risk around the LME and US government is a limitation on market appetite to short the metal. However, if that risk recedes, then we see significant pressure on nickel price emerging. Our latest 2022 and 2023 average price forecasts are \$24,300/t and \$18,500/t respectively.

**Exhibit 30: A bigger class 2 glut as EU stainless steel demand faces growing headwinds**



Source: Goldman Sachs Global Investment Research

**Exhibit 31: Nickel supply likely to remain strong until 2025**



Source: Woodmac, Goldman Sachs Global Investment Research

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Exhibit 32: GS Global Nickel Supply and Demand Model

| ('000 tonnes)                       | 2020        | 2021        | 2022E       | 2023E       | 2024E       | 2025E       |
|-------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>Consumption - DM</b>             |             |             |             |             |             |             |
| US                                  | 136         | 132         | 141         | 158         | 192         | 247         |
| % change y/y                        | -13%        | -3%         | 7%          | 12%         | 22%         | 28%         |
| Europe                              | 282         | 339         | 316         | 325         | 376         | 417         |
| % change y/y                        | -9%         | 20%         | -7%         | 3%          | 16%         | 11%         |
| Japan                               | 143         | 165         | 173         | 187         | 193         | 204         |
| % change y/y                        | -5%         | 15%         | 5%          | 8%          | 3%          | 5%          |
| Other DM                            | 7           | 8           | 9           | 9           | 9           | 9           |
| % change y/y                        | -11%        | 7%          | 8%          | 2%          | 2%          | 2%          |
| <b>Sub- DM</b>                      | <b>568</b>  | <b>644</b>  | <b>639</b>  | <b>678</b>  | <b>770</b>  | <b>877</b>  |
| % change y/y                        | -9%         | 13%         | -1%         | 6%          | 14%         | 14%         |
| <b>Consumption - EM</b>             |             |             |             |             |             |             |
| China                               | 1447        | 1525        | 1639        | 1779        | 1855        | 1909        |
| Indonesia                           | 206         | 383         | 395         | 442         | 521         | 547         |
| China & Indonesia                   | 1653        | 1908        | 2034        | 2221        | 2376        | 2456        |
| % change y/y                        | 10%         | 15%         | 7%          | 9%          | 7%          | 3%          |
| Other EM                            | 266         | 300         | 272         | 285         | 295         | 310         |
| % change y/y                        | -18%        | 13%         | -9%         | 5%          | 4%          | 5%          |
| <b>Sub- EM</b>                      | <b>1919</b> | <b>2207</b> | <b>2306</b> | <b>2506</b> | <b>2671</b> | <b>2766</b> |
| % change y/y                        | 5%          | 15%         | 4%          | 9%          | 7%          | 4%          |
| <b>Global Consumption</b>           | <b>2487</b> | <b>2851</b> | <b>2945</b> | <b>3184</b> | <b>3441</b> | <b>3643</b> |
| % change y/y                        | 1%          | 15%         | 3%          | 8%          | 8%          | 6%          |
| World ex-China & Indonesia          | 832         | 943         | 933         | 989         | 1075        | 1178        |
| % change y/y                        | -12%        | 13%         | -1%         | 6%          | 9%          | 10%         |
| <b>Global Production</b>            |             |             |             |             |             |             |
| <b>Global Primary Production</b>    | <b>2541</b> | <b>2708</b> | <b>3003</b> | <b>3257</b> | <b>3478</b> | <b>3699</b> |
| % change y/y                        | 7%          | 7%          | 11%         | 8%          | 7%          | 6%          |
| China                               | 780         | 760         | 830         | 890         | 996         | 1055        |
| % change y/y                        | -4%         | -3%         | 9%          | 7%          | 12%         | 6%          |
| Indonesia                           | 638         | 895         | 1123        | 1247        | 1299        | 1431        |
| % change y/y                        | 65%         | 40%         | 26%         | 11%         | 4%          | 10%         |
| ex-China & Indonesia                | 1123        | 1053        | 1050        | 1120        | 1183        | 1214        |
| % change y/y                        | -4%         | -6%         | 0%          | 7%          | 6%          | 3%          |
| <b>Global Battery Scrap Supply</b>  | <b>3</b>    | <b>7</b>    | <b>10</b>   | <b>19</b>   | <b>32</b>   | <b>51</b>   |
| <b>Global Balance</b>               | <b>52</b>   | <b>-136</b> | <b>68</b>   | <b>92</b>   | <b>69</b>   | <b>108</b>  |
| <b>Cash Prices (annual average)</b> |             |             |             |             |             |             |
| Current dollars (\$/t)              | 13803       | 18474       | 24343       | 18500       | 18000       | 18000       |
| Current dollars (c/lb)              | 626         | 838         | 1104        | 839         | 816         | 816         |

Source: Goldman Sachs Global Investment Research

## Appendix

Exhibit 33: GS lithium &amp; cobalt price forecasts

| Unit:\$/t | Lithium carbonate<br>China spot (ex VAT) | Lithium hydroxide<br>Asia CIF | Spodumene 6%<br>China CIF | Cobalt metal<br>standard grade | Cobalt hydroxide<br>China index | Cobalt hydroxide<br>payable (%) |
|-----------|------------------------------------------|-------------------------------|---------------------------|--------------------------------|---------------------------------|---------------------------------|
| 2022E     | 62929                                    | 47150                         | 4331                      | 69100                          | 51134                           | 74                              |
| 2023E     | 53304                                    | 46793                         | 4330                      | 48375                          | 31444                           | 65                              |
| 2024E     | 11000                                    | 12500                         | 800                       | 44000                          | 27720                           | 63                              |

For Lithium carbonate China spot, we take out a 13% VAT post conversion from Yuan/tonne

Source: Goldman Sachs Global Investment Research

Exhibit 34: GS Cobalt Green Demand Model

| GS Cobalt Green Demand Model      |           |           |           |           |            |            |            |            |            |            |
|-----------------------------------|-----------|-----------|-----------|-----------|------------|------------|------------|------------|------------|------------|
| 000 tonnes                        | 2021      | 2022E     | 2023E     | 2024E     | 2025E      | 2026E      | 2027E      | 2028E      | 2029E      | 2030E      |
| <b>Europe</b>                     | <b>8</b>  | <b>7</b>  | <b>9</b>  | <b>11</b> | <b>14</b>  | <b>19</b>  | <b>23</b>  | <b>27</b>  | <b>33</b>  | <b>38</b>  |
| EV                                | 8         | 7         | 8         | 10        | 13         | 17         | 21         | 25         | 31         | 37         |
| ESS                               | 0         | 0         | 0         | 1         | 1          | 1          | 1          | 2          | 2          | 2          |
| YoY growth                        | 96%       | -9%       | 23%       | 25%       | 28%        | 33%        | 22%        | 18%        | 24%        | 16%        |
| <b>US</b>                         | <b>4</b>  | <b>4</b>  | <b>7</b>  | <b>11</b> | <b>17</b>  | <b>26</b>  | <b>28</b>  | <b>30</b>  | <b>35</b>  | <b>40</b>  |
| EV                                | 4         | 4         | 7         | 10        | 16         | 24         | 26         | 28         | 33         | 38         |
| ESS                               | 0         | 0         | 0         | 1         | 1          | 2          | 2          | 2          | 2          | 2          |
| YoY growth                        | 54%       | 1%        | 81%       | 53%       | 65%        | 49%        | 9%         | 5%         | 18%        | 14%        |
| <b>China</b>                      | <b>30</b> | <b>42</b> | <b>55</b> | <b>62</b> | <b>67</b>  | <b>72</b>  | <b>74</b>  | <b>74</b>  | <b>86</b>  | <b>99</b>  |
| EV                                | 29        | 39        | 52        | 58        | 63         | 68         | 69         | 69         | 82         | 95         |
| ESS                               | 1         | 2         | 3         | 4         | 5          | 5          | 5          | 4          | 4          | 4          |
| YoY growth                        | 162%      | 38%       | 32%       | 13%       | 9%         | 8%         | 2%         | 0%         | 17%        | 15%        |
| <b>RoW</b>                        | <b>4</b>  | <b>3</b>  | <b>4</b>  | <b>5</b>  | <b>5</b>   | <b>5</b>   | <b>6</b>   | <b>8</b>   | <b>12</b>  | <b>15</b>  |
| EV                                | 4         | 3         | 4         | 4         | 4          | 5          | 6          | 8          | 12         | 14         |
| ESS                               | 0         | 0         | 0         | 0         | 0          | 0          | 0          | 0          | 1          | 1          |
| YoY growth                        | 50%       | -12%      | 20%       | 18%       | 1%         | 10%        | 27%        | 30%        | 44%        | 25%        |
| <b>Global cobalt green demand</b> | <b>48</b> | <b>58</b> | <b>78</b> | <b>93</b> | <b>108</b> | <b>128</b> | <b>138</b> | <b>146</b> | <b>174</b> | <b>202</b> |
| EV                                | 44        | 53        | 70        | 82        | 96         | 114        | 122        | 130        | 158        | 184        |
| ESS                               | 2         | 3         | 4         | 6         | 7          | 8          | 9          | 8          | 8          | 8          |
| E-buses, two-wheeler EVs          | 3         | 2         | 3         | 5         | 5          | 6          | 7          | 8          | 8          | 10         |
| YoY growth                        | 116%      | 20%       | 34%       | 19%       | 17%        | 18%        | 7%         | 6%         | 19%        | 16%        |

Source: Goldman Sachs Global Investment Research

Exhibit 35: GS Lithium Green Demand Model

| GS Lithium Green Demand Model      |            |            |            |            |             |             |             |             |             |             |
|------------------------------------|------------|------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|
| '000 tonnes LCE                    | 2021       | 2022E      | 2023E      | 2024E      | 2025E       | 2026E       | 2027E       | 2028E       | 2029E       | 2030E       |
| <b>Europe</b>                      | <b>40</b>  | <b>51</b>  | <b>67</b>  | <b>91</b>  | <b>125</b>  | <b>174</b>  | <b>226</b>  | <b>295</b>  | <b>366</b>  | <b>450</b>  |
| EV                                 | 37         | 47         | 61         | 82         | 113         | 157         | 203         | 266         | 332         | 413         |
| ESS                                | 3          | 4          | 6          | 9          | 13          | 17          | 23          | 29          | 33          | 37          |
| YoY growth                         | 102%       | 26%        | 31%        | 37%        | 38%         | 39%         | 30%         | 31%         | 24%         | 23%         |
| <b>US</b>                          | <b>20</b>  | <b>27</b>  | <b>53</b>  | <b>89</b>  | <b>158</b>  | <b>243</b>  | <b>282</b>  | <b>327</b>  | <b>385</b>  | <b>466</b>  |
| EV                                 | 18         | 25         | 48         | 80         | 142         | 219         | 253         | 295         | 350         | 427         |
| ESS                                | 2          | 2          | 5          | 9          | 16          | 24          | 28          | 32          | 35          | 38          |
| YoY growth                         | 59%        | 40%        | 93%        | 67%        | 78%         | 54%         | 16%         | 16%         | 18%         | 21%         |
| <b>China</b>                       | <b>155</b> | <b>298</b> | <b>420</b> | <b>518</b> | <b>606</b>  | <b>678</b>  | <b>735</b>  | <b>816</b>  | <b>954</b>  | <b>1164</b> |
| EV                                 | 143        | 274        | 382        | 467        | 545         | 611         | 661         | 736         | 867         | 1069        |
| ESS                                | 12         | 24         | 37         | 50         | 61          | 67          | 74          | 79          | 87          | 95          |
| YoY growth                         | 171%       | 92%        | 41%        | 23%        | 17%         | 12%         | 8%          | 11%         | 17%         | 22%         |
| <b>RoW</b>                         | <b>19</b>  | <b>23</b>  | <b>30</b>  | <b>38</b>  | <b>42</b>   | <b>48</b>   | <b>65</b>   | <b>94</b>   | <b>135</b>  | <b>178</b>  |
| EV                                 | 17         | 21         | 27         | 35         | 38          | 43          | 58          | 85          | 122         | 163         |
| ESS                                | 1          | 2          | 3          | 4          | 4           | 5           | 6           | 9           | 12          | 15          |
| YoY growth                         | 55%        | 22%        | 28%        | 29%        | 9%          | 15%         | 35%         | 45%         | 44%         | 32%         |
| <b>Global lithium green demand</b> | <b>267</b> | <b>441</b> | <b>634</b> | <b>821</b> | <b>1033</b> | <b>1262</b> | <b>1453</b> | <b>1718</b> | <b>2074</b> | <b>2520</b> |
| EV                                 | 215        | 367        | 518        | 664        | 837         | 1030        | 1176        | 1382        | 1671        | 2073        |
| ESS                                | 18         | 32         | 51         | 72         | 94          | 113         | 131         | 149         | 168         | 185         |
| E-buses, two-wheeler EVs           | 34         | 42         | 66         | 86         | 102         | 120         | 146         | 187         | 235         | 263         |
| YoY growth                         | 120%       | 65%        | 44%        | 29%        | 26%         | 22%         | 15%         | 18%         | 21%         | 22%         |

Source: Goldman Sachs Global Investment Research

Exhibit 36: GS Nickel Green Demand Model

| GS Nickel Green Demand Model      |            |            |            |            |            |            |             |             |             |             |
|-----------------------------------|------------|------------|------------|------------|------------|------------|-------------|-------------|-------------|-------------|
| 000 tonnes                        | 2021       | 2022E      | 2023E      | 2024E      | 2025E      | 2026E      | 2027E       | 2028E       | 2029E       | 2030E       |
| <b>Europe</b>                     | <b>39</b>  | <b>40</b>  | <b>46</b>  | <b>68</b>  | <b>91</b>  | <b>130</b> | <b>163</b>  | <b>206</b>  | <b>251</b>  | <b>291</b>  |
| EV                                | 32         | 32         | 37         | 59         | 79         | 115        | 143         | 183         | 226         | 265         |
| ESS                               | 1          | 1          | 2          | 2          | 4          | 5          | 6           | 8           | 9           | 10          |
| Wind                              | 6          | 7          | 7          | 7          | 8          | 10         | 13          | 15          | 15          | 16          |
| YoY growth                        | 73%        | 3%         | 13%        | 49%        | 34%        | 42%        | 26%         | 26%         | 22%         | 16%         |
| <b>US</b>                         | <b>22</b>  | <b>21</b>  | <b>33</b>  | <b>63</b>  | <b>108</b> | <b>171</b> | <b>190</b>  | <b>216</b>  | <b>250</b>  | <b>288</b>  |
| EV                                | 16         | 17         | 30         | 57         | 100        | 160        | 179         | 204         | 238         | 274         |
| ESS                               | 0          | 1          | 1          | 2          | 5          | 7          | 8           | 9           | 10          | 10          |
| Wind                              | 6          | 3          | 3          | 3          | 4          | 3          | 3           | 3           | 3           | 3           |
| YoY growth                        | 33%        | -4%        | 60%        | 88%        | 72%        | 58%        | 11%         | 14%         | 16%         | 15%         |
| <b>China</b>                      | <b>141</b> | <b>214</b> | <b>271</b> | <b>378</b> | <b>438</b> | <b>505</b> | <b>527</b>  | <b>569</b>  | <b>653</b>  | <b>750</b>  |
| EV                                | 123        | 187        | 235        | 333        | 383        | 447        | 467         | 508         | 590         | 686         |
| ESS                               | 4          | 7          | 10         | 14         | 18         | 19         | 21          | 22          | 24          | 26          |
| Wind                              | 15         | 21         | 26         | 32         | 38         | 39         | 39          | 39          | 39          | 39          |
| YoY growth                        | 83%        | 52%        | 26%        | 40%        | 16%        | 15%        | 4%          | 8%          | 15%         | 15%         |
| <b>RoW</b>                        | <b>23</b>  | <b>22</b>  | <b>26</b>  | <b>33</b>  | <b>36</b>  | <b>41</b>  | <b>51</b>   | <b>69</b>   | <b>95</b>   | <b>118</b>  |
| EV                                | 15         | 15         | 17         | 25         | 26         | 32         | 41          | 58          | 83          | 105         |
| ESS                               | 0          | 1          | 1          | 1          | 1          | 1          | 2           | 3           | 3           | 4           |
| Wind                              | 8          | 7          | 9          | 8          | 8          | 8          | 8           | 8           | 8           | 10          |
| YoY growth                        | 57%        | -5%        | 16%        | 28%        | 8%         | 14%        | 25%         | 35%         | 38%         | 24%         |
| <b>Global green demand nickel</b> | <b>244</b> | <b>327</b> | <b>415</b> | <b>589</b> | <b>731</b> | <b>909</b> | <b>1005</b> | <b>1143</b> | <b>1336</b> | <b>1546</b> |
| EV                                | 186        | 251        | 318        | 473        | 588        | 753        | 831         | 953         | 1137        | 1330        |
| ESS                               | 5          | 9          | 14         | 20         | 27         | 32         | 37          | 42          | 47          | 50          |
| E-buses, two-wheeler EVs          | 18         | 29         | 40         | 47         | 58         | 62         | 74          | 82          | 86          | 99          |
| Wind                              | 35         | 38         | 43         | 50         | 58         | 61         | 63          | 65          | 65          | 68          |
| YoY growth                        | 74%        | 34%        | 27%        | 42%        | 24%        | 24%        | 11%         | 14%         | 17%         | 16%         |

Source: Goldman Sachs Global Investment Research

## Exhibit 37: GS Cobalt Recycling Supply Model

| GS Cobalt Recycling Supply Model    |          |          |          |           |           |           |           |           |           |           |
|-------------------------------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| '000 tonnes                         | 2021     | 2022E    | 2023E    | 2024E     | 2025E     | 2026E     | 2027E     | 2028E     | 2029E     | 2030E     |
| <b>Global EOL batteries</b>         | <b>0</b> | <b>0</b> | <b>2</b> | <b>4</b>  | <b>8</b>  | <b>10</b> | <b>11</b> | <b>14</b> | <b>26</b> | <b>32</b> |
| Europe                              | 0        | 0        | 0        | 1         | 1         | 2         | 2         | 3         | 5         | 6         |
| US                                  | 0        | 0        | 0        | 1         | 1         | 2         | 2         | 3         | 5         | 7         |
| China                               | 0        | 0        | 2        | 3         | 5         | 6         | 6         | 7         | 13        | 16        |
| Japan                               | 0        | 0        | 0        | 0         | 0         | 0         | 0         | 0         | 1         | 1         |
| Korea                               | 0        | 0        | 0        | 0         | 0         | 0         | 0         | 1         | 1         | 2         |
| <b>Global production scrap</b>      | <b>4</b> | <b>4</b> | <b>5</b> | <b>6</b>  | <b>7</b>  | <b>8</b>  | <b>8</b>  | <b>9</b>  | <b>10</b> | <b>11</b> |
| <b>Global total recycled supply</b> | <b>4</b> | <b>5</b> | <b>7</b> | <b>10</b> | <b>15</b> | <b>18</b> | <b>19</b> | <b>23</b> | <b>36</b> | <b>43</b> |

Source: Goldman Sachs Global Investment Research

## Exhibit 38: GS Lithium Recycling Supply Model

| GS Lithium Recycling Supply Model   |          |           |           |           |           |           |           |            |            |            |
|-------------------------------------|----------|-----------|-----------|-----------|-----------|-----------|-----------|------------|------------|------------|
| '000 tonnes LCE                     | 2021     | 2022E     | 2023E     | 2024E     | 2025E     | 2026E     | 2027E     | 2028E      | 2029E      | 2030E      |
| <b>Global EOL batteries</b>         | <b>0</b> | <b>1</b>  | <b>10</b> | <b>21</b> | <b>38</b> | <b>49</b> | <b>50</b> | <b>66</b>  | <b>129</b> | <b>231</b> |
| Europe                              | 0        | 0         | 1         | 3         | 5         | 7         | 9         | 13         | 26         | 46         |
| US                                  | 0        | 0         | 1         | 2         | 6         | 10        | 11        | 14         | 27         | 48         |
| China                               | 0        | 1         | 8         | 15        | 24        | 29        | 28        | 35         | 67         | 119        |
| Japan                               | 0        | 0         | 0         | 0         | 1         | 1         | 1         | 1          | 3          | 6          |
| Korea                               | 0        | 0         | 0         | 1         | 1         | 1         | 2         | 3          | 6          | 13         |
| <b>Global production scrap</b>      | <b>9</b> | <b>15</b> | <b>21</b> | <b>27</b> | <b>33</b> | <b>41</b> | <b>47</b> | <b>55</b>  | <b>67</b>  | <b>83</b>  |
| <b>Global total recycled supply</b> | <b>9</b> | <b>16</b> | <b>31</b> | <b>47</b> | <b>71</b> | <b>90</b> | <b>97</b> | <b>121</b> | <b>196</b> | <b>314</b> |

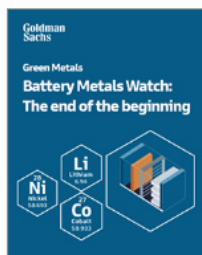
Source: Goldman Sachs Global Investment Research

## Exhibit 39: GS Nickel Recycling Supply Model

| GS Nickel Recycling Supply Model    |          |           |           |           |           |           |           |           |            |            |
|-------------------------------------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| '000 tonnes                         | 2021     | 2022E     | 2023E     | 2024E     | 2025E     | 2026E     | 2027E     | 2028E     | 2029E      | 2030E      |
| <b>Global EOL batteries</b>         | <b>0</b> | <b>1</b>  | <b>8</b>  | <b>15</b> | <b>30</b> | <b>40</b> | <b>43</b> | <b>61</b> | <b>110</b> | <b>149</b> |
| Europe                              | 0        | 0         | 1         | 2         | 4         | 6         | 8         | 12        | 22         | 30         |
| US                                  | 0        | 0         | 1         | 2         | 5         | 9         | 9         | 13        | 23         | 31         |
| China                               | 0        | 1         | 6         | 11        | 20        | 24        | 24        | 32        | 57         | 77         |
| Japan                               | 0        | 0         | 0         | 0         | 1         | 1         | 1         | 1         | 3          | 4          |
| Korea                               | 0        | 0         | 0         | 0         | 1         | 1         | 1         | 2         | 5          | 8          |
| <b>Global production scrap</b>      | <b>7</b> | <b>9</b>  | <b>11</b> | <b>17</b> | <b>21</b> | <b>26</b> | <b>29</b> | <b>33</b> | <b>40</b>  | <b>47</b>  |
| <b>Global total recycled supply</b> | <b>7</b> | <b>10</b> | <b>19</b> | <b>32</b> | <b>51</b> | <b>67</b> | <b>73</b> | <b>94</b> | <b>150</b> | <b>195</b> |

Source: Goldman Sachs Global Investment Research

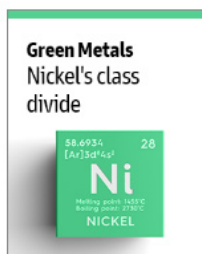
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29 May 2022

Battery metals – cobalt, lithium and nickel – will power the green industrial revolution, facing a wave of demand comparable to that of copper and iron ore during China's rapid growth in the 2000's. With climate change top of mind, investors are fully aware that battery metals will play a crucial role in the 21st century global economy, just as bulk and base metals did before them. Yet despite this exponential demand profile, we see the battery metals bull market as over for now.



### Green Metals: Nickel's class divide

28 April 2022

At the beginning of the year, nickel's place within green metals was as a key competitor in the race for mineral dominance of energy storage. Now, it sits at the intersection of Europe's push for decarbonisation and energy independence. At the heart of Europe's strategy lies its desire to rapidly electrify its transportation sector – a source of 20% of its emissions and c.2mb/d of Russian oil imports.



### Green Metals: Solving Aluminium's Climate Paradox

20 June 2021

At the heart of the coming surge in green aluminium demand lies a paradox: aluminium is a key input required to produce decarbonising technologies like EV's and solar power, yet its own production is very carbon intensive, generating 2% of all global emissions. This paradox begs the question: how can we secure enough aluminium to effectively decarbonise, while keeping the climate impact of the path to net zero to a minimum? In our view, the resolution of this paradox will drive a structural bull market in aluminium over the next half decade, driven by the necessity to grow supply to meet green demand while cutting emissions to prevent a climate catastrophe.



### Green Metals: Copper is the new oil

13 Apr 2021

The critical role copper will play in achieving the Paris climate goals cannot be overstated. As the most cost-effective conductive material, copper sits at the heart of capturing, storing and transporting these new sources of energy. Leveraging equity analysis across EVs, wind, solar, and battery technology, we estimate that by 2030, copper demand will grow nearly 600%. Yet the copper market as it currently stands is not prepared for this demand, and producers are leery of committing to new supply. We significantly increase our price forecasts and explore the implications across the metals and mining complex.

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We, Aditi Rai, Nicholas Snowdon, Annalisa Schiavon and Jeffrey Currie, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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